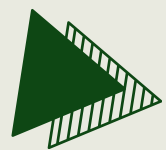


RAIL-PEDESTRIAN RISK

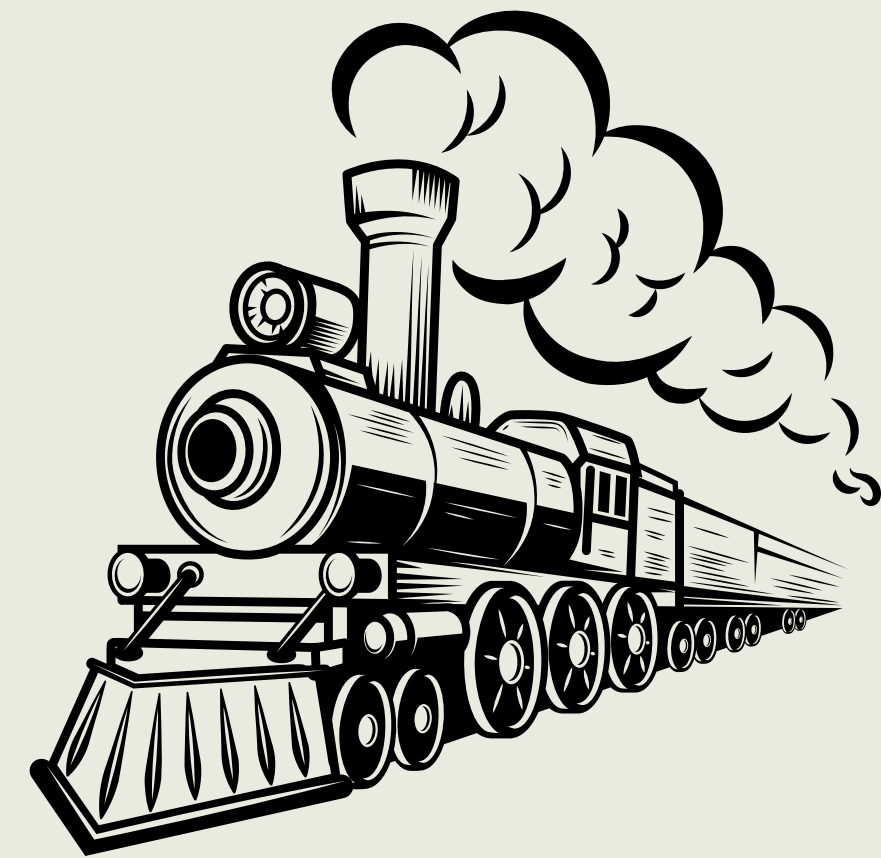
Understanding Pedestrian Risk at
Blocked Train Crossings



Anna Fetter
PLAN372,
SPRING 2025

Research Question

How do neighborhood walkability and demographic characteristics relate to pedestrian climbing incidents at blocked train crossings?



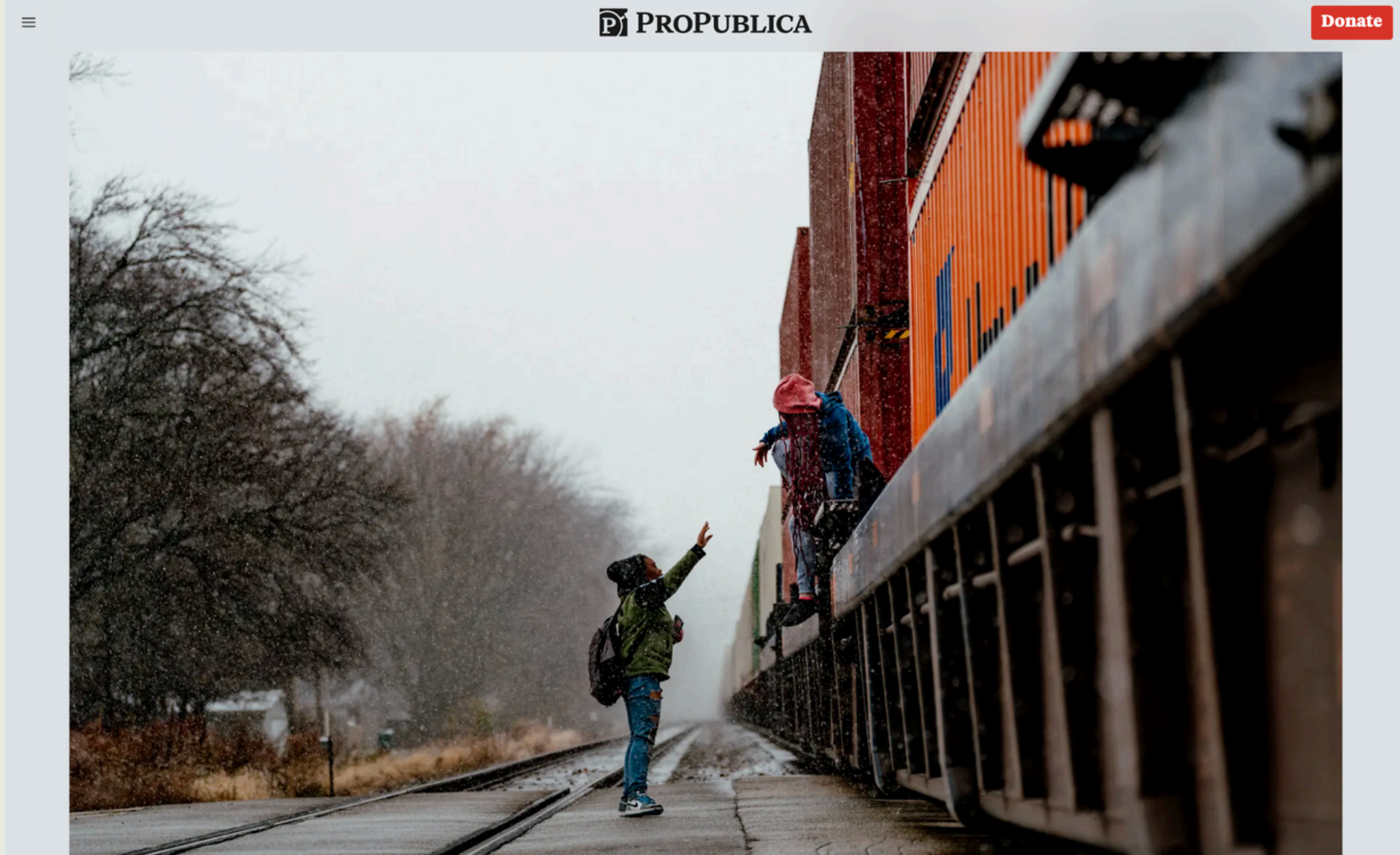
Motivation

- 76,000 blocked crossing complaints (2022-2024)
- 23% of complaints cited pedestrians climbing over, under, or through the blocked train crossing
- Preliminary data analysis showed clusters in certain cities or states
 - Houston, TX had 7/10 most blocked crossings
- Civilian reported data

As Rail Profits Soar, Blocked Crossings Force Kids to Crawl Under Trains to Get to School

by Topher Sanders and Dan Schwartz, ProPublica, and Jocie Sterman, Gray Television/InvestigateTV; Video by Scotty Smith, Gray Television/InvestigateTV; Photography by Jamie Kelter Davis for ProPublica

April 26, 2023, 12 p.m. EDT



One child helps another cross over a parked freight train blocking their route to school.

Data Sources

FRA blocked crossings database (2022-2024)

<https://www.fra.dot.gov/blockedcrossings/incidents>

joined with the location of the crossingIDs

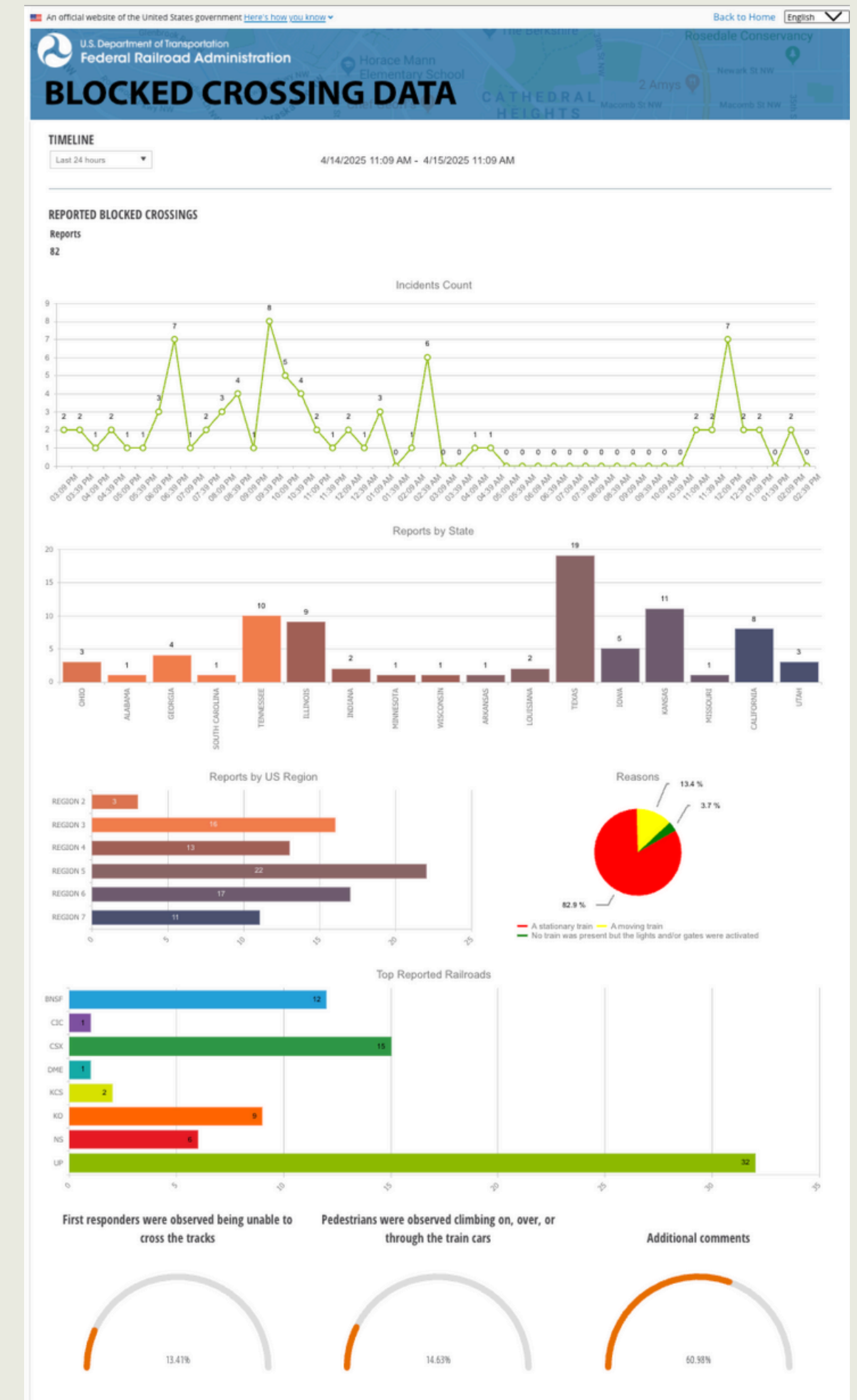
<https://data.transportation.gov/Railroads/Crossing-Inventory-Data-Form-71-Current>

EPA Walkability Index (block group)

<https://www.epa.gov/smartgrowth/smart-location-mapping#walkability>

ACS demographics (block & tract level)

Used tidycensus in R



Methods

Data cleaning & spatial joins

I made sure everything I needed had appropriate spatial data. I added walkability and census data to the blocked crossing complaints. Because of processing constraints, I focused on the 5 states with the most complaints (~half of total complaints).

Prepare data for regression

I aggregated complaints by crossing to calculate metrics like the percentage of complaints involving pedestrians climbing or long delays. I also added in walkability, population density, and median income columns.

Multivariable linear regression

I modeled climbing percentage using predictors like walkability, income, population density, percent of households without cars, and if the delay exceeded 15 minutes.

Identify high risk crossings

I created a risk score for each crossing using the factors in the regression. This allowed me to identify and map the top five high-risk crossings.

Regression Results

```
Call:
lm(formula = climb_pct ~ walkability + pop_density + median_income +
    pct_no_vehicle + long_delay_pct, data = crossing_summary_5_states)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.46876	-0.13978	-0.03875	0.09732	0.67400

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.356e-01	5.517e-02	-2.457	0.014190	*
walkability	1.512e-02	2.054e-03	7.357	4.43e-13	***
pop_density	2.367e-05	7.631e-06	3.102	0.001984	**
median_income	-9.694e-07	2.916e-07	-3.324	0.000926	***
pct_no_vehicle	1.629e-03	1.155e-03	1.410	0.158930	
long_delay_pct	2.772e-01	4.792e-02	5.784	1.02e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.201 on 849 degrees of freedom
(246 observations deleted due to missingness)

Multiple R-squared: 0.2183, Adjusted R-squared: 0.2137

F-statistic: 47.42 on 5 and 849 DF, p-value: < 2.2e-16

meaningful patterns: longer delays, higher walkability, and lower median income are consistently associated with more climbing incidents

249 observations were deleted due to missing (census) data

relatively weak R^2 , only explains 22% of variation, too low to use for forecasting

High Risk Crossings

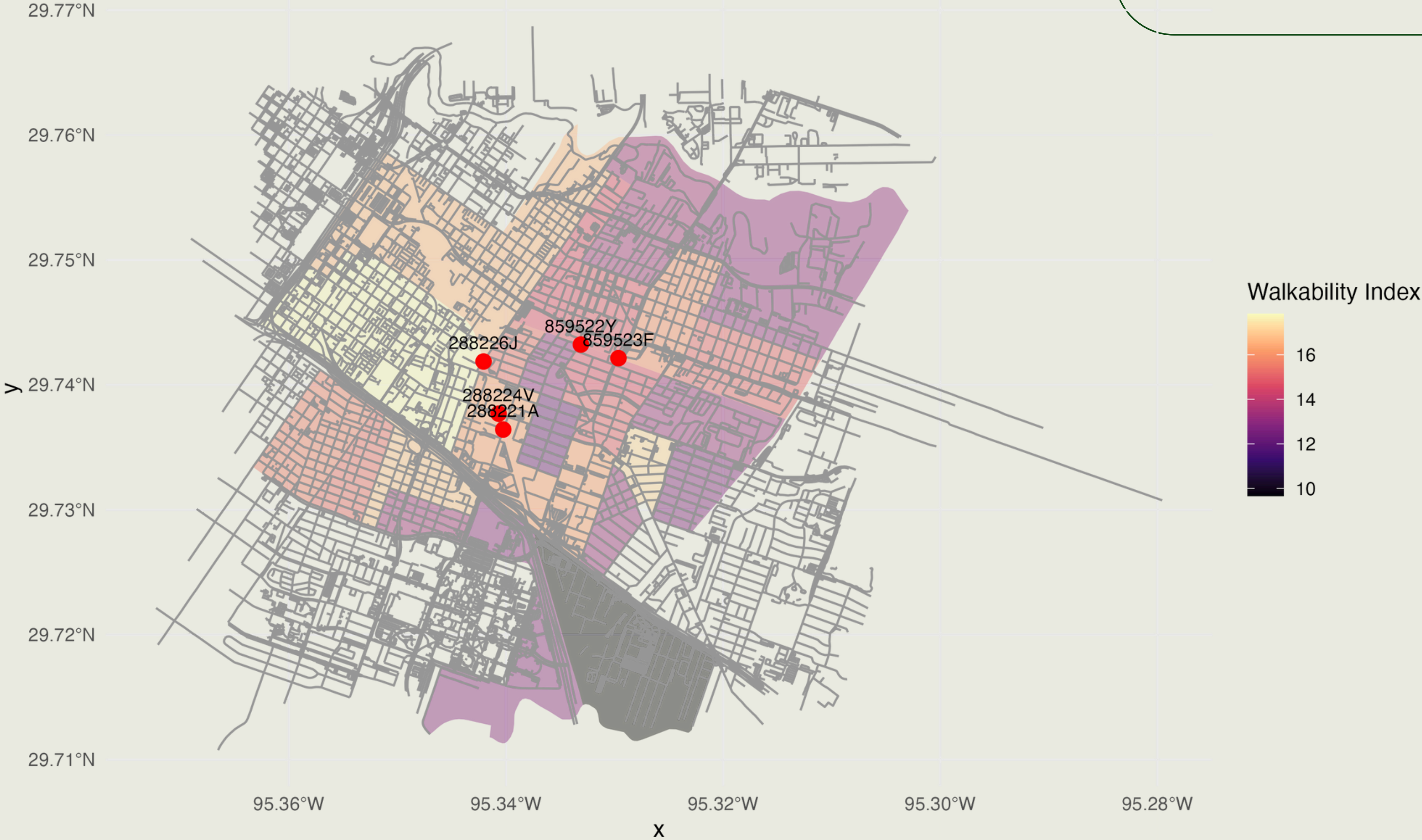
I created a composite risk score to identify the most dangerous crossings by combining climbing behavior, delay severity, walkability, complaint volume, and income data. Variables are standardized and weighted based on their expected contribution to risk.

```
high_risk_crossings <- crossing_summary_5_states %>%
  filter(total_complaints >= 5) %>%
  mutate(
    risk_score = (
      scale(climb_pct)[, 1] * 1.5 +           # Weighted highest bc it's the main outcome of concerns
      scale(walkability)[, 1] * 1 +
      scale(long_delay_pct)[, 1] * 1 +
      scale(total_complaints)[, 1] * 1.2 -    # Elevate frequent complaints at a crossing
      scale(median_income)[, 1] * 0.5        # Lower-income vulnerability
    )
  ) %>%
  arrange(desc(risk_score)) %>%
  slice_head(n = 20)
```

This is not a predictive tool. Use only to identify existing high risk crossings.

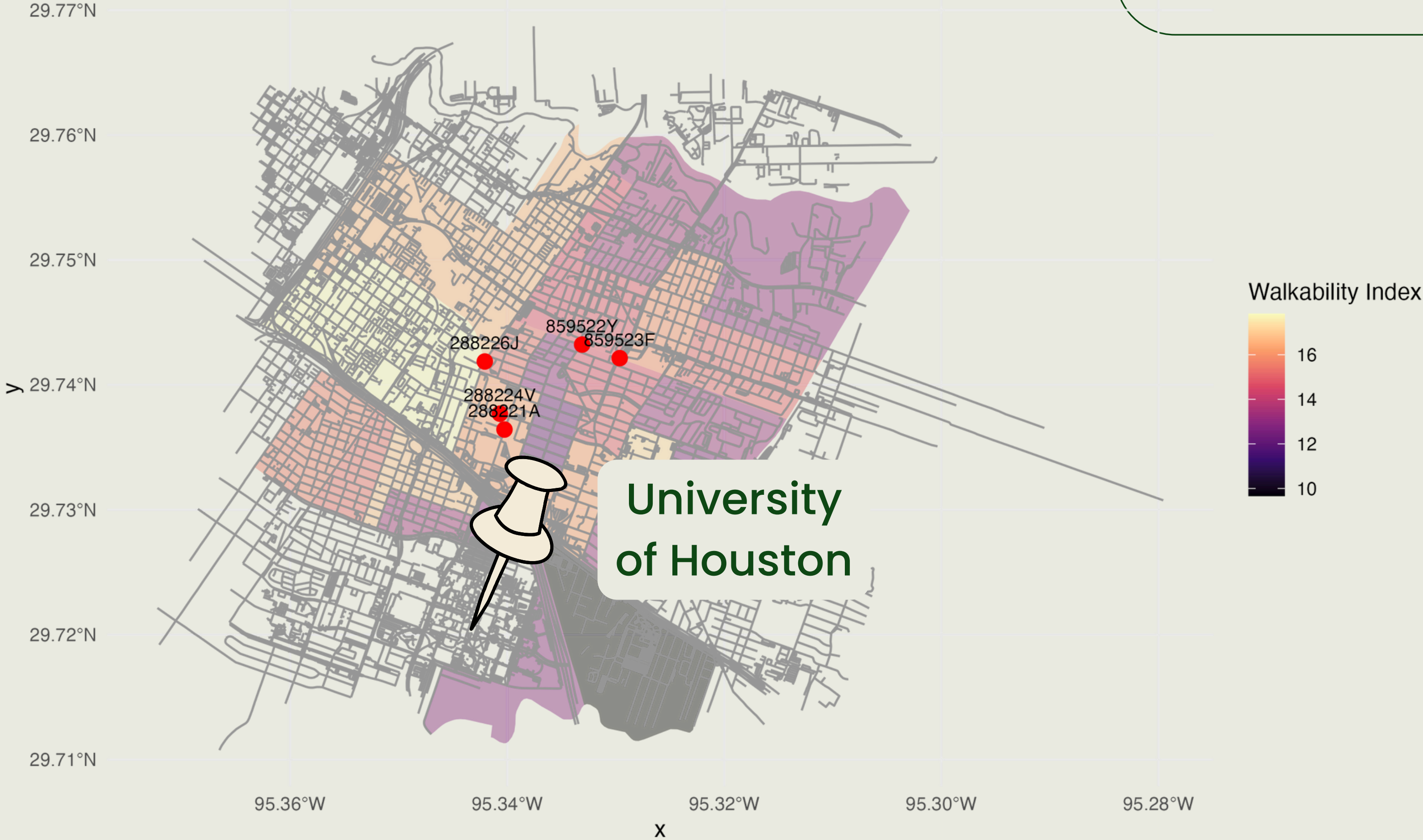
High-Risk Train Crossings in Houston, TX

High Risk Crossings



High-Risk Train Crossings in Houston, TX

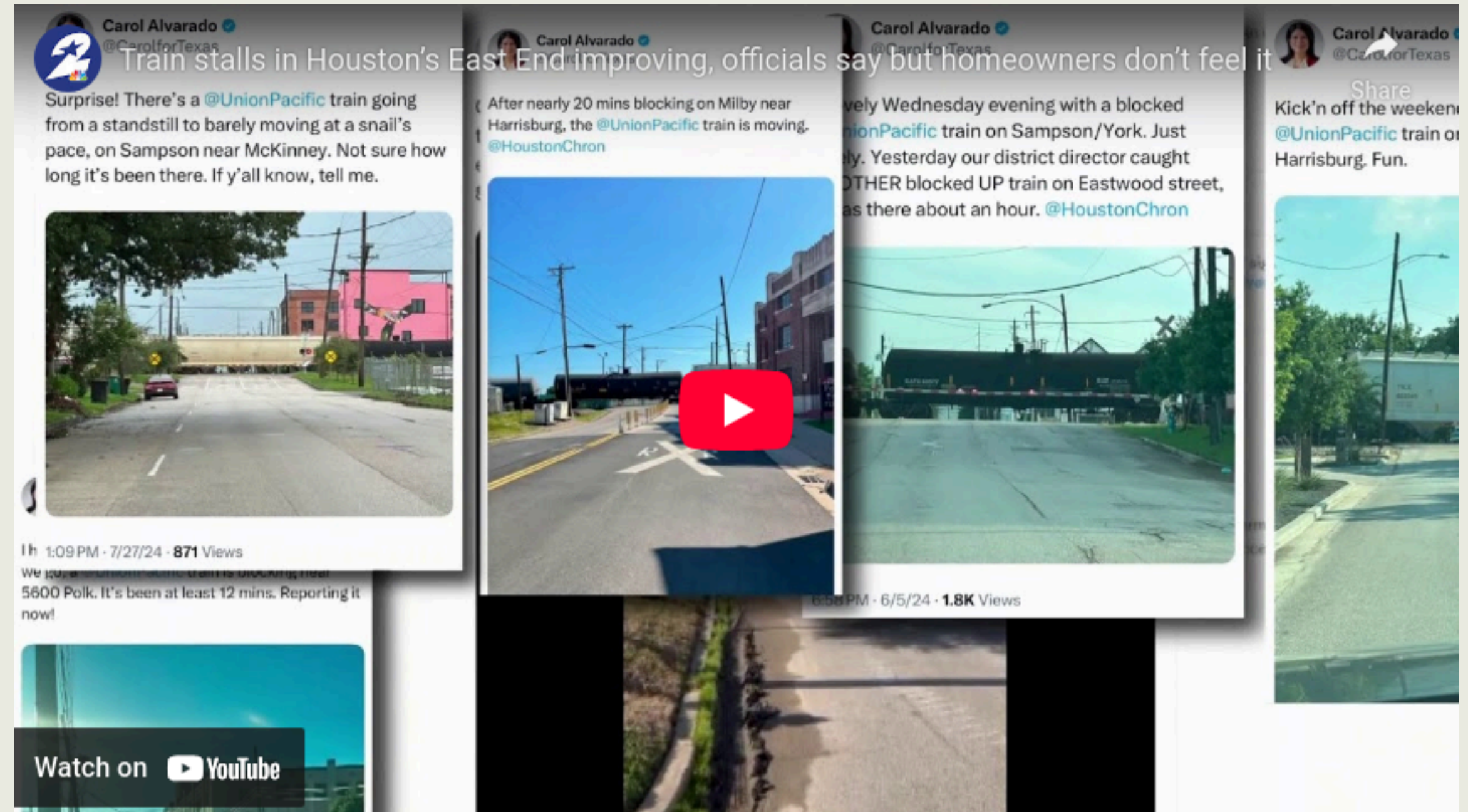
High Risk Crossings



Policy Implications

Add data behind what residents in East Houston are already experiencing

Railroads are under federal regulation, so much of the policy change & funding for overpasses/underpasses need to come from federal sources.



References / Repo

<https://github.com/aefetter/plan372-blocked-train-crossings>

<https://www.propublica.org/article/trains-crossing-blocked-kids-norfolk-southern>

<https://www.epa.gov/smartgrowth/smart-location-mapping#walkability>

<https://data.transportation.gov/Railroads/Crossing-Inventory-Data-Form-71-Current/>

<https://www.fra.dot.gov/blockedcrossings/incidents>